1 A ball is dropped on to an inclined thin metal sheet, as shown in Fig. 1.1.

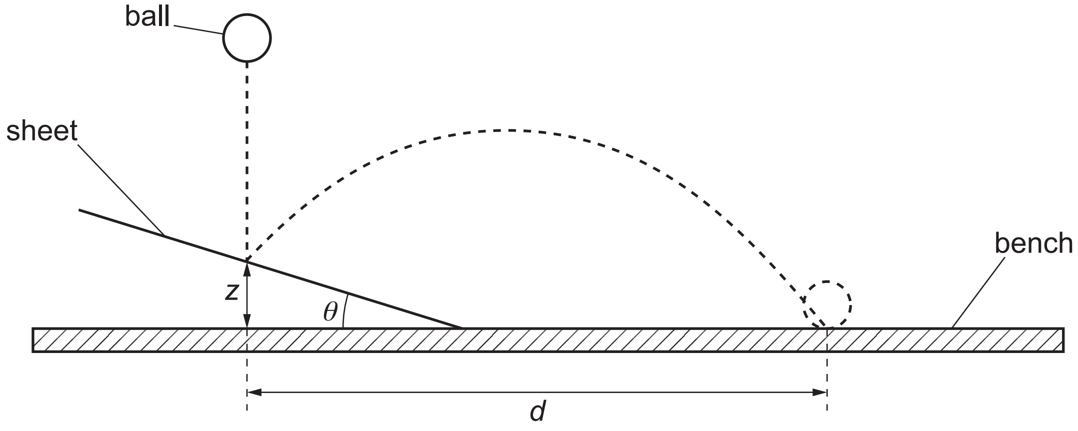

**Fig. 1.1** (not to scale)

The angle between the sheet and the horizontal bench is $\theta$. The height of the point of contact of the ball and the sheet is $z$. The horizontal distance travelled by the ball between its points of contact with the sheet and the bench is $d$, as shown in Fig. 1.1.

It is suggested that $d$ is related to $\theta$ by the relationship

$$d = \frac{Pv^2 \sin 4\theta}{g} + Q\sqrt{z}$$

where $v$ is the speed of the ball as it makes contact with the sheet, $g$ is the acceleration of free fall, and $P$ and $Q$ are constants.

Plan a laboratory experiment to test the relationship between $d$ and $\theta$.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for $P$ and $Q$.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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**Diagram**

[15]

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